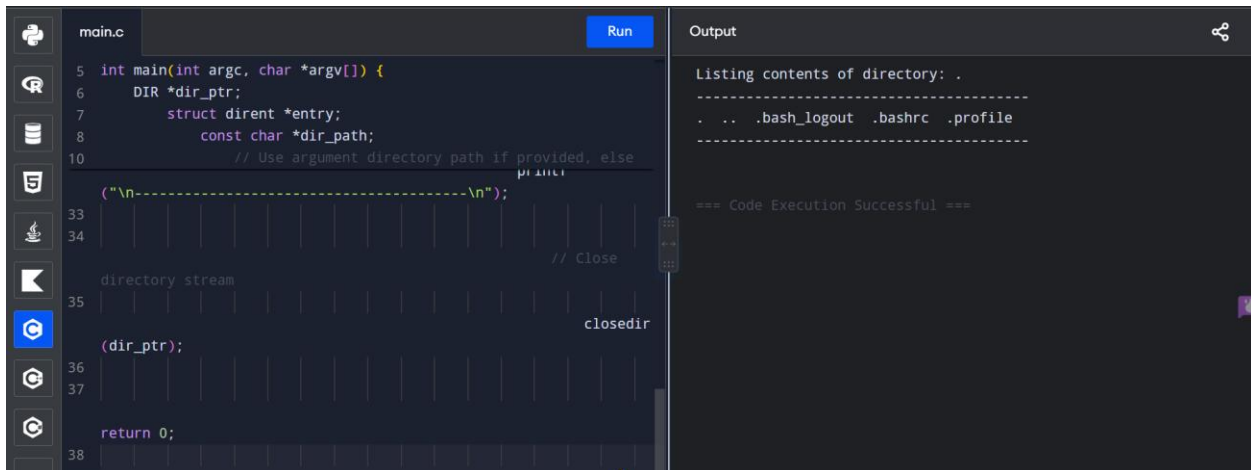


EXPERIMENT 37:

37. Develop a C program for simulating the function of ls UNIX Command.



The screenshot shows a code editor with a C program named `main.c` and its output. The code is as follows:

```
5 int main(int argc, char *argv[]) {
6     DIR *dir_ptr;
7     struct dirent *entry;
8     const char *dir_path;
9     // Use argument directory path if provided, else
10    // use current directory
11    if (argc > 1)
12        dir_path = argv[1];
13    else
14        dir_path = ".";
15    printf("\n-----\n");
16    // Open directory stream
17    dir_ptr = opendir(dir_path);
18    if (dir_ptr == NULL) {
19        perror("opendir");
20        return 1;
21    }
22    // Loop through directory entries
23    while ((entry = readdir(dir_ptr)) != NULL) {
24        printf("%s\n", entry->d_name);
25    }
26    // Close directory stream
27    closedir(dir_ptr);
28    return 0;
29 }
```

The output of the program is:

```
Listing contents of directory: .
-----
.  .. .bash_logout .bashrc .profile
-----
=== Code Execution Successful ===
```

PSEUDO CODE :

- Check if a directory path argument was provided via command-line arguments (argc, argv). If provided, set target directory to argv[1]; otherwise, default to standard current directory ".".
- Open the directory stream using opendir().
- If directory pointer is NULL, print an error message using perror() and exit.
- Loop through the directory entries using readdir() inside a while loop.
- In each iteration, retrieve the filename string from entry->d_name and display it.
- Close the directory stream using closedir().

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <dirent.h>
```

```
int main(int argc, char
*argv[]) {

    DIR *dir_ptr;

    struct dirent *entry;

    const char *dir_path;


    // Use argument
    directory path if
    provided, else default to
    current directory "."

    if (argc > 1) {

        dir_path = argv[1];

    } else {

        dir_path = ".";

    }


    // Open directory
    stream

    dir_ptr =
```

```
opendir(dir_path);

    if (dir_ptr == NULL)
    {

        perror("Unable to
open directory");

        return 1;

    }
```

```
    printf("Listing
contents of directory:
%s\n", dir_path);

    printf("-----
-----\n");
```

```
    // Read and print
directory entries

    while ((entry =
readdir(dir_ptr)) !=
NULL) {
```

```
        // Print filename
```

```
        printf("%s ",
entry->d_name);

    }

    printf("\n-----
-----
\n");
```

```
    // Close directory
    stream

    closedir(dir_ptr);

    return 0;

}
```

INPUT :

./ls_sim

OUTPUT:

Listing contents of
directory: .

. .. main.c ls_sim

Makefile notes.txt

data.csv
